QPCODE: **Reg. No :……..**
Name : ……

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) DEGREE EXAMINATION
FOURTH SEMESTER
(2024 Admission Onwards)
Discipline Specific Elective (DSE) Course

MG4DSECHEA00: CHARACTERIZATION OF NANOMATERIALS

Time : 2hrs **Total Marks: 70**

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

Section A
(Answer any five Questions)

1. Distinguish different modes of AFM.[An] [CO-2]
2. Define Thermogravimetric Analysis (TGA) and explain its basic working principle [R] [CO-4]
3. Describe the fundamental principle of XPS?[U] [CO-3]
4. Discuss the role of DSC in determining the thermal stability of polymer-based nanocomposites. [A] [CO-4]
5. Explain the significances of BET analysis (U) [CO-4]
6. How can analyse the crystallite size using Jmol Software?(U) [CO-6]
7. Explain Polarography? Which is electrode is used in Polarography?[U] [CO-5]

(5X4=20 Marks)

Section B
(Answer any five Questions)

8. What is SEM? Explain the working principle of SEM? [U] [CO-1]
9. Explain the working principle of Transmission Electron Microscopy. [U] [CO-6]
10. Illustrate the working principle of STM.[An] [CO-2]
11. Describe how DSC is used to analyze glass transition, melting, and crystallization of nanomaterials with two examples. [A] [CO-4]
12. How can be done the analysis of CNTs and Graphene using Raman spectroscopy (An) [CO-4]
13. Explain the techniques used for the analysis of the variation of viscosity of nanocomposites when nanofillers are added into it? (U) [CO-4]
14. Differentiate between XRD and SAXS using for the analysis of nanomaterials (U) [CO-4]

(5X7=35 Marks)

Section C
(Answer any one question)

15. Discuss the Principle, working mechanism, basic Instrumental features of FESEM (U) [CO-1]
16. Write a note about (a) SERS and TERS and (b) Raman Spectroscopy for the analysis of nanomaterials(U) [CO-3]

(1X15=15 Marks)